Deepak Maurya

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Research Interests

I develop computationally efficient algorithms with near-minimax statistical guarantees for computationally intractable learning problems. My recent work focuses on high-dimensional sparse regression under adversarial and heavy-tailed corruption.

Education

2021 – Present **Purdue University**, **Ph.D. in Computer Science**, **GPA: 4.0/4.0**
Advisors: Jean Honorio and Petros Drineas. ***Anticipated Graduation: May 2027***

2018 – 2021 **Indian Institute of Technology Madras**, **M.S. in Computer Science & Engineering**
Thesis: Spectral Hypergraph Theory. Advisors: Balaraman Ravindran, Shankar Narasimhan.

2011 – 2016 **Indian Institute of Technology Madras**, **M.Tech & B.Tech (Dual Degree) in Electrical Engineering**
Thesis: Identification of Linear Dynamic Systems using Dynamic Iterative PCA. Advisors: Arun K. Tangirala, Shankar Narasimhan.

Selected Publications

1. Robust Estimation of a Sparse Linear Model: Provable Guarantees with Non-convexity, **Deepak Maurya**, Adarsh Barik, Jean Honorio, AISTATS 2026.
2. On Exact Solutions of the Inner Optimization Problem of Adversarial Robustness, **Deepak Maurya**, Adarsh Barik, Jean Honorio, ICASSP 2025, [Paper].
3. A Theoretical Study of the Effects of Adversarial Attacks on Sparse Regression, **Deepak Maurya**, Jean Honorio, TMLR 2024, [Paper].
4. HEAL: Unlocking the Potential of Learning on Hypergraphs Enriched with Attributes and Layers, Naganand Yadati, Tarun Kumar, **Deepak Maurya**, Partha Talukdar, Balaraman Ravindran, LoG 2023, [Paper].
5. Hypergraph Partitioning using Tensor Eigenvalue Decomposition, **Deepak Maurya**, Balaraman Ravindran, PLOS ONE 2023, [Paper].
6. Identification of Output-Error (OE) Models using Generalized Spectral Decomposition, **Deepak Maurya**, Arun K. Tangirala, Shankar Narasimhan, ICC 2019, **Best Student Paper Award**, [Paper], [Code], [Slides].

Papers Under Preparation/Submission

1. Provable Guarantees for Robust Feature Selection in Sparse Linear Models in High Dimensions, **Deepak Maurya**, Adarsh Barik, Jean Honorio.
2. One Eigenvector Solves Isomorphism For Nearly All Random Graphs, **Deepak Maurya**, Balaraman Ravindran, Srinivasan Parthasarathy.

Academic Service

2021 – 2026 Co-organized **five** editions of Graphs & more Complex structures for Learning & Reasoning (GCLR) workshop at AAAI (**A*** conference).

2024 – 2026 Reviewer for NeurIPS 2024-25, ICML 2025-26, AISTATS 2025-26, UAI 2026, ICASSP 2025.

2021 – 2026 Teaching Assistant at Purdue for CS 373, CS 578: Undergrad and graduate level machine learning courses. CS 182: Foundations of computer science

Awards

2023 – 2024 **Summer Research Grant** by Purdue Graduate School.

2019 **Best Student Paper** at Indian Control Conference (ICC) 2019.